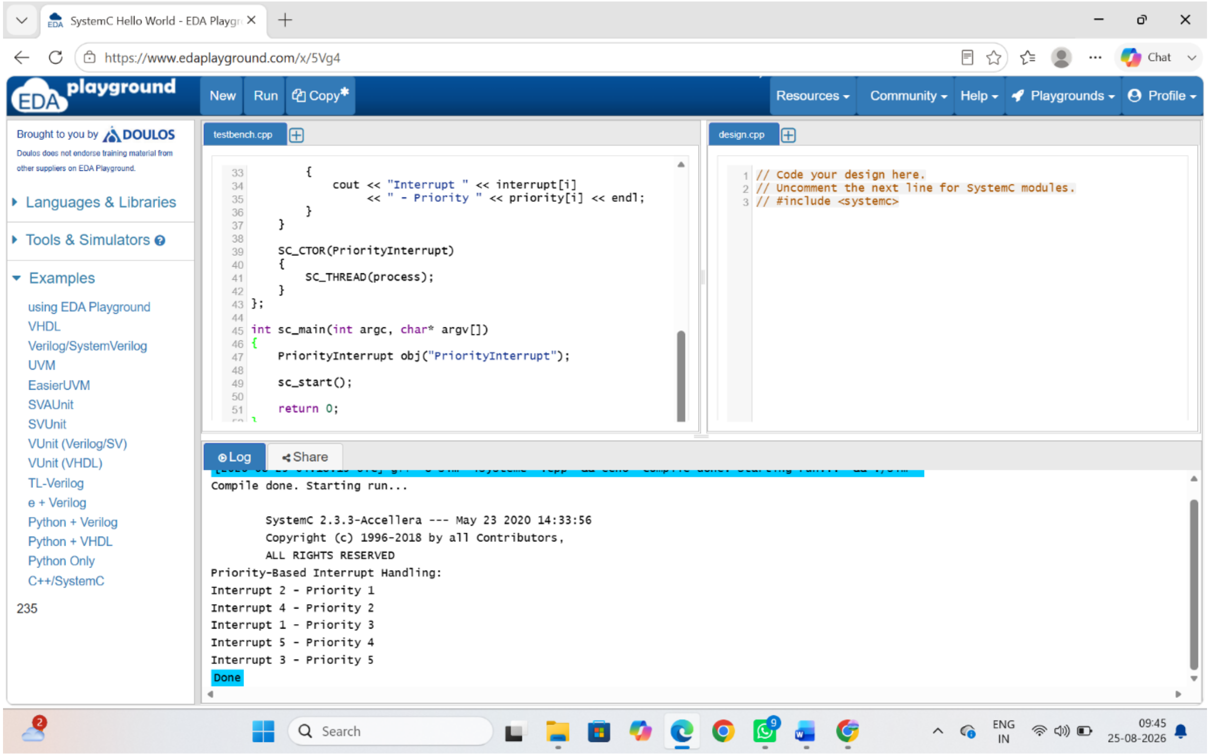
**EXP NO 29 Priority-Based Interrupt Handling**

**PROGRAM C++**

```
#include <systemc.h>
using namespace std;

SC_MODULE(PriorityInterrupt)
{
    void process()
    {
        int interrupt[5] = {1, 2, 3, 4, 5};
        int priority[5] = {3, 1, 5, 2, 4};
        int temp;

        // Sort according to priority
        for (int i = 0; i < 5; i++)
        {
            for (int j = i + 1; j < 5; j++)
            {
                if (priority[i] > priority[j])
                {
                    temp = priority[i];
                    priority[i] = priority[j];
                    priority[j] = temp;

                    temp = interrupt[i];
                    interrupt[i] = interrupt[j];
                    interrupt[j] = temp;
                }
            }
```

```
        }

        cout << "Priority-Based Interrupt Handling:" << endl;

        for (int i = 0; i < 5; i++)
        {
            cout << "Interrupt " << interrupt[i]
                 << " - Priority " << priority[i] << endl;
        }
    }

    SC_CTOR(PriorityInterrupt)
    {
        SC_THREAD(process);
    }
};

int sc_main(int argc, char* argv[])
{
    PriorityInterrupt obj("PriorityInterrupt");

    sc_start();

    return 0;
}
```

## OUTPUT




## PROGRAM C

```
#include <stdio.h>

int main()
{
    int interrupt[5] = {1, 2, 3, 4, 5};
    int priority[5] = {3, 1, 5, 2, 4};
    int i, j, temp;

    for (i = 0; i < 5; i++)
    {
        for (j = i + 1; j < 5; j++)
        {
            if (priority[i] > priority[j])
```

```c
            {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = interrupt[i];
                interrupt[i] = interrupt[j];
                interrupt[j] = temp;
            }
        }
    }

    printf("Priority-Based Interrupt Handling:\n");

    for (i = 0; i < 5; i++)
    {
        printf("Interrupt %d - Priority %d\n",
               interrupt[i], priority[i]);
    }

    return 0;
}
```

**OUTPUT**

**RESULT**

Both C and C++ programs successfully implement priority-based interrupt handling and produce the same interrupt execution order, with C using printf() and C++ using cout for output.